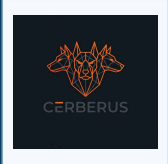


Options Skew Risk Reversal Predictor



INTERNAL

Use 25-delta risk reversal and skew as a directional predictor for SPX returns.

Author: Cerberus Research

Generated: 2026-08-07 08:52 UTC

N 1,840		EDGE +0.61%		HIT-RATE 100.0%	
CLASSIFICATION internal				AUTHOR Cerberus Research	
RUN ID 20260807_095207				GIT COMMIT 978cf61e	
DURATION 1.0s				DATASETS cboe/cboe_skew, cboe/cboe_smile	
PARAMETER · LOOKBACK_YEARS 10				PARAMETER · HORIZON 21	
PARAMETER · BENCHMARK SPY				KPI Value	
N 1,840				EDGE +0.61%	
HIT-RATE 100.0%					
KEY FINDINGS · Skew series loaded: price					

CONTENTS

- 01 Executive Summary
- 02 Results & Interpretation
- 03 Key Findings
- 04 Data Sources
- 05 Objectives
- 06 Methodology
- 07 Charts
- 08 Statistical Results
- 09 Limitations

Generated by the Cerberus studies framework · schema 2

Data Sources

- cboe/cboe_skew
- cboe/cboe_smile

Key Findings

- Skew series loaded: price

Executive Summary

NOTE

Tests options skew and risk-reversal signals as directional predictors for SPX returns.

Results & Interpretation

- Across 1,840 events, the forward return averaged +0.61%. Returns were positive 100.0% of the time.
- Across 1,840 events, the forward return averaged +0.61%. Returns were positive 97.9% of the time.
- Across 1,840 events, the forward return averaged +0.48%. Returns were positive 94.3% of the time.
- Across 1,839 events, the forward return averaged +0.66%. Returns were positive 89.8% of the time.
- Across 1,816 events, the forward return averaged +1.00%. Returns were positive 75.1% of the time.

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Analysis is run on the configured datasets with the parameters listed in the Study Configuration block. Results are aggregated into summary statistics, tables, and visual one-pagers for review.

Charts

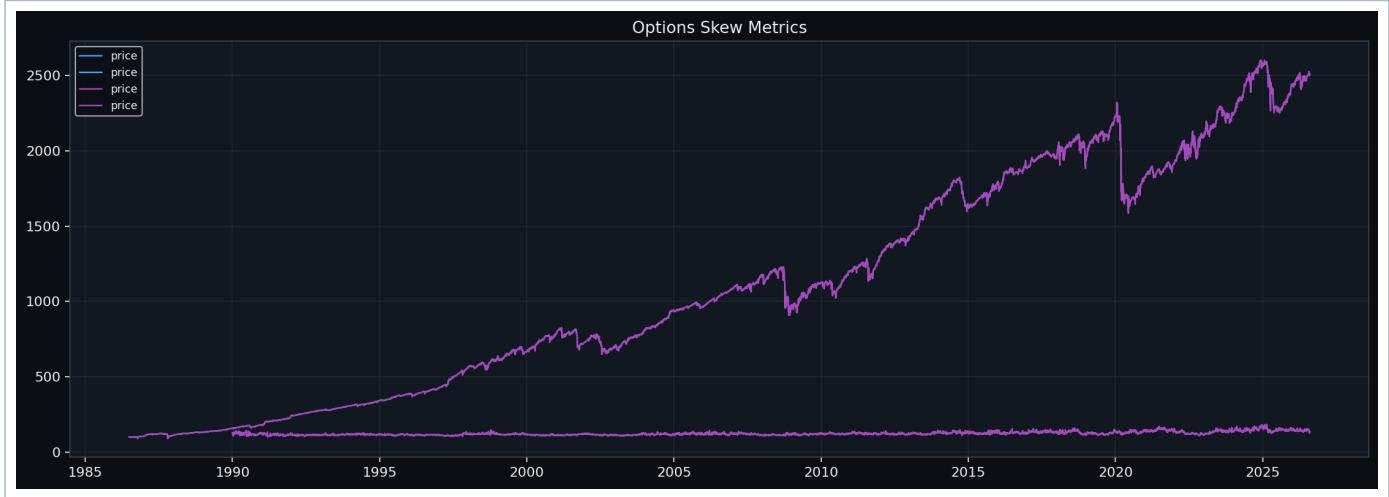


Figure 1. Options skew metrics
Options skew metrics

Statistical Results

price	price
149.71	2,267.96
147.03	2,268.23
147.53	2,271.65
148.55	2,270.73
151.45	2,275.55
153.13	2,278.66
150.62	2,276.72
153.79	2,281.13
153.95	2,281.54
153.17	2,275.74
144.56	2,273.19
144.32	2,268.88
146.39	2,285.20
149.99	2,282.29
156.53	2,285.83
150.88	2,288.07
152.50	2,291.71
152.10	2,285.63
151.48	2,280.14
150.22	2,285

price	price
153.96	2,291.84
148.16	2,290.05
148.82	2,292.56
150.93	2,293.89
151.92	2,295.91
147.63	2,302.50
149.49	2,300.82
152.11	2,307.43
148.21	2,305.75
146.37	2,301.59
148.19	2,304.67
146.97	2,316.68
145.46	2,323.04
146.53	2,317.01
145.87	2,313.69
145.89	2,309.13
141.51	2,314.81
143.03	2,317.95
143.44	2,320.69
143.68	2,324.51
145.93	2,325.11
144.65	2,324.55
142.50	2,328.45
145.22	2,325.02
145.76	2,328.90
145.34	2,326.07
138.86	2,308.28
142.81	2,321.59
146.48	2,321.43
144.98	2,323.44
147.14	2,324.11
147.63	2,330.53

price	price
151.31	2,335.61
149.95	2,338.53
152.85	2,341.73
154.34	2,342.48
153.49	2,338.33
148.50	2,326.68
140.37	2,321.54
139.45	2,320.80
141.15	2,336.60
142.78	2,334.74
141.99	2,337.46
144.01	2,349.22
143.64	2,350.67
143.54	2,357.49
141.27	2,360.36
142.94	2,353.45
143.45	2,353.45
138.57	2,353.66
142.49	2,366.05
137.93	2,369.51
140.90	2,373.43
142.98	2,374.43
143.05	2,376.23
144.96	2,376.52
148.34	2,377.54
148.60	2,379.12
151.68	2,375.77
145.04	2,369.99
143	2,366.15
144.86	2,374.91
144.94	2,375.97
149.23	2,372.83

price	price
148.91	2,373.60
148.83	2,372.88
150.63	2,380.14
150.32	2,383.56
147.55	2,373.87
148.32	2,372.89
153.59	2,396.05
161.86	2,401.67
156.99	2,410.08
153.49	2,422.06
155.38	2,422.02
155.87	2,424.81
159.93	2,432.06
160.53	2,437.57
161.30	2,442.37
151.45	2,441.92
150.47	2,437.92
148.33	2,436.25
148.70	2,428.85
141.86	2,430.75
142.45	2,436.71
146.23	2,442.58
143.57	2,438.74
147.08	2,438.18
146.04	2,443.06
149.75	2,444.83
151.60	2,440.58
152.52	2,435.05
151.57	2,434.22
153.59	2,433.10
154.96	2,422.88
144.41	2,435.46

price	price
141.31	2,440.44
144.02	2,441.85
148.10	2,444.78
147.25	2,444.79
144.98	2,445.13
143.74	2,446.85
143.65	2,447.67
146.56	2,452.97
143.85	2,451.78
143.16	2,448.36
137.16	2,437.91
140.31	2,457.96
145.23	2,462.33
145.71	2,463.20
143.78	2,464.79
142.50	2,455.20
139.35	2,459.13
140.28	2,465.79
139.17	2,470.38
141.19	2,471.75
145.52	2,472.01
141.43	2,470.72
141.94	2,477.52
146.05	2,481.19
146.57	2,484.13
146.67	2,482.97
151.78	2,483.75
152.83	2,479.26
152.87	2,487.69
153.95	2,484.57
151.80	2,468.49
157.98	2,485.69

price	price
154.46	2,485.70
152.44	2,489.49
139.94	2,472.81
137.76	2,469.65
141.49	2,492.74
145.04	2,500.45
136.54	2,485.15
138.26	2,485.06
139.12	2,476.21
142.02	2,483.31
142.63	2,485.32
140.88	2,489.92
143.99	2,486.88
139	2,476.22
142.23	2,482.96
145.45	2,506.04
143.77	2,508.92
146.95	2,513.68
147.60	2,518.15
148.45	2,513.79
150.36	2,504.82
143.27	2,501.87
144.18	2,505.75
156.93	2,491.21
149.94	2,462.60
139.23	2,442.67
140.74	2,436.98
141.82	2,417.56
141.90	2,415.42
140.91	2,406.33
140.84	2,418.72
139.59	2,413.98

price	price
139.08	2,423.28
139.64	2,426.25
138.16	2,421.35
141.88	2,420.76
143.33	2,430.34
141.38	2,433.47
141.74	2,429.53
138.74	2,436.66
135.42	2,456.46
136.11	2,450.14
138.21	2,462.69
140.21	2,467.18
139.41	2,462.28
141.51	2,474.76
139.32	2,493.10
145.77	2,469.42
138.40	2,469.53
135.50	2,462.81
132.31	2,474.55
136.96	2,477.03
137.39	2,481.05
139.04	2,489.09
138.57	2,488.73
139.52	2,494.52
144.18	2,494.89
142.46	2,498.04
143.24	2,499.05
136.86	2,491.05
142.15	2,493.82
152.25	2,473.31
145	2,478.36
141.97	2,475.92

Research only. Not investment advice. Past performance is not indicative of future results.

price	price
143.08	2,464.51
142.98	2,479.58
142.60	2,483.68
144.32	2,487.71
142.66	2,487.08
142.62	2,487.06
146.72	2,488.51
141.85	2,484.98
143.14	2,468.60
145.30	2,468.72
139.89	2,471.26
139.40	2,464.13
144.46	2,484.40
149.60	2,491.38
154.82	2,491.38
150.02	2,491.04
145.38	2,497.61
145.74	2,495.09
149.79	2,494.34
144.67	2,498.13
144.27	2,499.60
145.69	2,499.17
145.13	2,500.24
148.51	2,500.66
145.72	2,500.94
147.28	2,499.65
146.05	2,504.23
151.66	2,507.89
150.19	2,509.82
145.95	2,507.04
147.28	2,510.18
146.60	2,513.31

Research only. Not investment advice. Past performance is not indicative of future results.

price	price
142.98	2,515.90
139.55	2,505.70
139.90	2,523.12
141.23	2,526.34
139.96	2,523.91
126.41	2,499.30
133.32	2,505.57
134.73	2,511.44

Table 1. Options skew data

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q1	1840	+ 0.61%	+ 0.49%	+ 100.00%
Q2	1840	+ 0.61%	+ 0.49%	+ 97.93%
Q3	1840	+ 0.48%	+ 0.49%	+ 94.29%
Q4	1839	+ 0.66%	+ 0.49%	+ 89.83%
Q5	1816	+ 1.00%	+ 0.93%	+ 75.11%

Table 2. Forward returns by skew regime

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

regime_fwd_returns

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q1	1840	+ 0.61%	+ 0.49%	+ 100.00%
Q2	1840	+ 0.61%	+ 0.49%	+ 97.93%
Q3	1840	+ 0.48%	+ 0.49%	+ 94.29%
Q4	1839	+ 0.66%	+ 0.49%	+ 89.83%
Q5	1816	+ 1.00%	+ 0.93%	+ 75.11%

skew

date	price	price.1
------	-------	---------

date	price	price.1
2025-08-06	149.71	2,267.96
2025-08-07	147.03	2,268.23
2025-08-08	147.53	2,271.65
2025-08-11	148.55	2,270.73
2025-08-12	151.45	2,275.55
2025-08-13	153.13	2,278.66
2025-08-14	150.62	2,276.72
2025-08-15	153.79	2,281.13
2025-08-18	153.95	2,281.54
2025-08-19	153.17	2,275.74
2025-08-20	144.56	2,273.19
2025-08-21	144.32	2,268.88
2025-08-22	146.39	2,285.20
2025-08-25	149.99	2,282.29
2025-08-26	156.53	2,285.83

Statistics Summary

Key	Value
study	options_skew_risk_reversal_predictor
category	options
status	success
parameters	{'lookback_years': 10, 'horizon': 21, 'benchmark': 'SPY'}
datasets	['cboe/cboe_skew', 'cboe/cboe_smile']
rows	19298
git_commit	978cf61eefd9b125a5087073e70ef1c9d91f687b
executed_at	2026-08-07T08:52:07.693151+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.

Research only. Not investment advice. Past performance is not indicative of future results.

Term	Definition
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-07 08:52 UTC
Author	Cerberus Research
Study slug	options_skew_risk_reversal_predictor
Run ID	20260807_095207
Git commit	978cf61e
Datasets	cboe/cboe_skew, cboe/cboe_smile
Parameters hash	ae4d5d218d7d
Parameter · lookback_years	10
Parameter · horizon	21
Parameter · benchmark	SPY
Data fingerprint	cbbffb58cdc2
Duration	1.0s